WORKED SOLUTION

# STEP II 2008, Question 2

Pure • Partial Fractions and Series

**STEP II 2008, Question 2**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s2-q2

## The key idea

Partial fractions turn one awkward rational function into three pieces whose expansions we know by heart: a geometric series, its derivative, and a geometric series in $-x^2$. The first two give a coefficient growing like $n$; the third only ever contributes $\pm\frac{1}{2}$, repeating with period 4, and that is where the "either/or" comes from. For the last part, look for a value of $x$ that turns the function into the given fraction.

## Solution

### The partial fractions

Since $(1-x)^2$ is a repeated linear factor and $1+x^2$ is an irreducible quadratic, we write

$$\frac{1+x}{(1-x)^2(1+x^2)} = \frac{A}{1-x} + \frac{B}{(1-x)^2} + \frac{Cx+D}{1+x^2},$$

so that, for all $x$,

$$1+x = A(1-x)(1+x^2) + B(1+x^2) + (Cx+D)(1-x)^2.$$

- Putting $x=1$: $2 = 2B$, so $B=1$.
- Putting $x=0$: $1 = A+B+D$, so $A=-D$.
- Putting $x=\mathrm{i}$ (allowed, since this is a polynomial identity): $1+\mathrm{i} = (C\mathrm{i}+D)(1-\mathrm{i})^2 = (C\mathrm{i}+D)(-2\mathrm{i}) = 2C-2D\mathrm{i}$. Comparing real and imaginary parts, $C=\frac{1}{2}$ and $D=-\frac{1}{2}$, hence $A=\frac{1}{2}$.

As a check, the coefficient of $x^3$ on the right is $-A+C=0$, matching the left. So

$$\frac{1+x}{(1-x)^2(1+x^2)} = \frac{1}{2(1-x)} + \frac{1}{(1-x)^2} - \frac{1-x}{2(1+x^2)}.$$

### The coefficients

For $|x| < 1$ all three expansions converge, and

$$\frac{1}{2(1-x)} = \tfrac{1}{2}\sum_{n\geq 0} x^n, \qquad \frac{1}{(1-x)^2} = \sum_{n\geq 0}(n+1)x^n,$$

$$\begin{aligned} -\frac{1-x}{2(1+x^2)} &= -\tfrac{1}{2}(1-x)\left(1-x^2+x^4-x^6+\cdots\right) \\ &= -\tfrac{1}{2}\left(1-x-x^2+x^3+x^4-x^5-x^6+x^7+\cdots\right). \end{aligned}$$

The bracket in the last line has coefficients $1, -1, -1, 1$ repeating with period 4: for $n = 2k$ the coefficient of $x^n$ is $(-1)^k$, and for $n = 2k+1$ it is $-(-1)^k$. So the third piece contributes $-\frac{1}{2}$ to $x^n$ when $n \equiv 0$ or $3 \pmod 4$, and $+\frac{1}{2}$ when $n \equiv 1$ or $2 \pmod 4$.

**Ask yourself** Why does the period-4 pattern produce exactly two cases rather than four?

Because only the sign matters: adding the three contributions,

$$a_n = \tfrac{1}{2} + (n+1) \mp \tfrac{1}{2} = \begin{cases} n+1, & n \equiv 0 \text{ or } 3 \pmod 4, \\ n+2, & n \equiv 1 \text{ or } 2 \pmod 4, \end{cases}$$

which is the required result. The sequence begins $1, 3, 4, 4, 5, 7, 8, 8, 9, 11, 12, 12, \ldots$

### The decimal approximation

**Ask yourself** Which value of $x$ makes the function equal to $\frac{11\,000}{8181}$?

The denominators suggest $x = \frac{1}{10}$: then $(1-x)^2 = 0.81$ and $1 + x^2 = 1.01$, so $(1-x)^2(1+x^2) = 0.8181$, and

$$\frac{1+\frac{1}{10}}{0.8181} = \frac{1.1}{0.8181} = \frac{11\,000}{8181}.$$

Since $\frac{1}{10} < 1$, the series is valid, and

$$\frac{11\,000}{8181} = \sum_{n\geq 0}\frac{a_n}{10^n} = 1 + 0.3 + 0.04 + 0.004 + 0.0005 + 0.00007 + \cdots.$$

While $a_n \leq 9$ (that is, up to $n = 8$) the coefficients are simply the decimal digits: 1.34457889. After that the coefficients have two digits, so we must carry:

$$\begin{aligned} \sum_{n=0}^{8}\frac{a_n}{10^n} &= 1.344\,578\,89 \\ +\,11\times 10^{-9} &\to 1.344\,578\,901 \\ +\,12\times 10^{-10} &\to 1.344\,578\,902\,2 \\ +\,12\times 10^{-11} &\to 1.344\,578\,902\,32 \\ +\,13\times 10^{-12} &\to 1.344\,578\,902\,333. \end{aligned}$$

The later terms only change the eleventh significant figure onwards, so to nine significant figures

$$\frac{11\,000}{8181} \approx 1.344\,578\,90.$$

## Checks

- Setting $x = 0$ in the function gives $1 = a_0$, and $n = 0$ falls in the $n + 1$ case. Multiplying out $(1 + x)(1 + 2x + \cdots)(1 - x^2 + \cdots)$ gives $1 + 3x + \cdots$, so $a_1 = 3 = 1 + 2$.
- Long division gives $11\,000/8181 = 1.344\,578\,902\,33\ldots$, agreeing with the series. We also checked $a_n$ for $n < 60$ by exact power-series division, and the partial fractions identity at many random rational points.

## Common slips

- Forgetting the carries: reading the digits straight off the $a_n$ gives $1.34457889\ldots$, which is wrong in the ninth significant figure.
- Writing $\frac{D}{1+x^2}$ instead of $\frac{Cx+D}{1+x^2}$; the irreducible quadratic needs a linear numerator.
- Rounding $1.344578902\ldots$ to $1.34457890$ and then dropping the final zero; it is a significant figure here.

## Beyond STEP

A rational function always has a power series whose coefficients are a polynomial in $n$ plus terms like $r^n$ and periodic patterns, one for each root of the denominator; this is the theory of *linear recurrences* and *generating functions*. Here the roots 1 (twice) and $\pm$i give the linear growth and the period-4 wobble. It is also why every fraction has an eventually repeating decimal. Wilf's *generatingfunctionology* is a lively and freely available introduction, and Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

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